

BOURG ST-MAURICE, France

WMO: 074970

Lat: 45.6126N

Lon: 6.7633E

Elev: 868

StdP: 91.32

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.0	-8.0	-14.3	1.2	-4.3	-12.4	1.4	-4.0	8.6	6.0	7.3	6.4	0.7	350	0.488

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	31.6	17.9	29.7	17.6	27.9	17.1	19.1	28.5	18.3	27.2	17.6	25.9	2.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	12.8	20.8	15.4	12.2	20.3	14.7	11.6	19.8	58.2	28.7	55.4	27.2	52.9	26.3	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.1	5.9	4.5	-12.7	34.2	3.3	2.0	-15.1	35.7	-17.0	36.8	-18.8	37.9	-21.2	39.4		
			-13.3	21.0	3.2	0.7	-15.6	21.5	-17.4	21.9	-19.2	22.3	-21.6	22.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	0.8	2.2	6.5	10.0	13.9	17.9	19.7	19.1	15.3	11.4	5.4	1.5
	DBStd	7.59	3.43	3.91	3.73	3.69	3.56	3.68	3.35	3.31	3.30	3.40	4.20	3.68
	HDD10.0	1114	284	219	118	44	8	1	0	0	3	26	147	264
	HDD18.3	3086	542	452	368	249	142	51	25	30	100	216	389	522
	CDD10.0	1244	0	0	9	45	130	238	300	283	162	69	8	1
	CDD18.3	173	0	0	0	0	5	38	67	54	9	0	0	0
	CDH23.3	1988	0	0	0	8	96	471	713	574	121	5	0	0
	CDH26.7	658	0	0	0	0	17	153	253	212	22	0	0	0
(14) Wind	WSAvg	1.5	1.2	1.4	1.7	2.0	1.7	1.5	1.4	1.3	1.3	1.4	1.5	1.3
(15) Precipitation	PrecAvg	1244	106	96	90	89	114	113	109	99	93	99	116	120
	PrecMax	1580	294	359	279	161	221	199	185	184	195	237	305	360
	PrecMin	952	12	14	20	10	43	64	41	29	32	19	33	0
	PrecStd	158	73	77	66	40	47	38	36	35	41	53	68	85
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	15.8	20.9	24.4	28.9	33.3	34.1	34.1	29.3	23.8	18.0	12.8
		MCWB	5.9	6.8	9.2	11.9	15.9	17.9	18.0	18.1	17.0	13.7	9.8	6.4
	2%	DB	9.5	12.6	18.5	21.8	26.1	30.5	31.6	31.4	26.7	21.3	15.5	10.9
		MCWB	4.4	5.5	8.4	10.9	14.5	17.8	18.0	17.9	16.2	13.0	8.6	5.4
	5%	DB	7.5	10.1	16.1	19.7	23.5	28.3	29.5	29.0	24.4	19.3	13.8	8.5
		MCWB	3.1	4.3	7.5	9.9	13.5	17.2	17.6	17.6	15.2	12.1	7.8	4.2
	10%	DB	5.6	8.0	13.5	17.6	21.4	26.1	27.5	26.7	22.1	17.4	12.0	6.4
		MCWB	2.4	3.1	6.3	9.0	12.5	16.2	17.1	16.9	14.4	11.3	7.0	3.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.8	7.7	10.2	12.6	17.0	20.0	20.2	20.1	18.2	15.3	10.9	7.3
		MCDB	10.0	13.6	18.2	22.7	26.3	29.6	29.6	29.8	25.9	20.9	15.1	10.9
	2%	WB	5.1	6.0	8.9	11.4	15.5	18.7	19.0	19.0	16.9	14.1	9.3	6.1
		MCDB	8.0	11.2	16.4	20.5	23.9	28.4	28.6	28.0	25.0	19.0	13.8	9.4
	5%	WB	3.9	4.8	7.9	10.4	14.2	17.7	18.2	18.1	15.9	13.0	8.3	5.1
		MCDB	6.6	9.3	14.8	18.4	21.4	26.7	27.3	26.4	22.7	17.6	12.4	7.6
	10%	WB	2.8	3.6	6.9	9.4	13.1	16.7	17.5	17.3	14.9	12.0	7.4	3.9
		MCDB	5.0	7.2	12.4	16.4	19.9	24.7	25.8	24.8	20.6	16.2	11.2	6.0
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	9.8	11.7	11.6	12.0	13.2	13.4	13.0	12.2	11.2	8.7	7.5
		MCDBR	9.8	13.4	16.0	15.7	16.1	16.8	16.7	17.1	15.9	14.0	11.4	9.7
		MCWBR	6.1	7.6	7.6	6.8	6.5	6.7	6.0	6.2	6.6	6.9	5.9	5.8
	5% WB	MCDBR	8.6	12.1	14.2	14.5	14.5	15.5	15.0	15.1	14.6	12.5	10.3	7.8
		MCWBR	5.7	7.0	7.0	6.5	6.3	6.6	6.1	6.1	6.4	6.5	5.7	5.2
(40) Clear-Sky Solar Irradiance	taub	0.269	0.300	0.355	0.406	0.421	0.421	0.411	0.396	0.376	0.350	0.307	0.270	(40)
	taud	2.298	2.212	2.120	2.101	2.148	2.208	2.243	2.293	2.337	2.379	2.400	2.353	(41)
	Ebn at Noon	839	872	863	849	848	849	852	851	838	807	787	799	(42)
	Edh at Noon	78	104	133	149	147	140	133	122	107	89	72	67	(43)
(44) All-Sky Solar Radiation	RadAvg	1.62	2.48	3.56	4.41	5.05	6.04	6.16	5.37	4.16	2.70	1.61	1.35	(44)
	RadStd	0.18	0.31	0.45	0.54	0.43	0.47	0.46	0.35	0.32	0.31	0.24	0.18	(45)

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (73 neighbors)	N/A	N/A	N/A	+0.72	+0.34	+0.23	N/A	N/A	+60	+20	(47)

Nomenclature: See separate page